

Turbocharger — LH

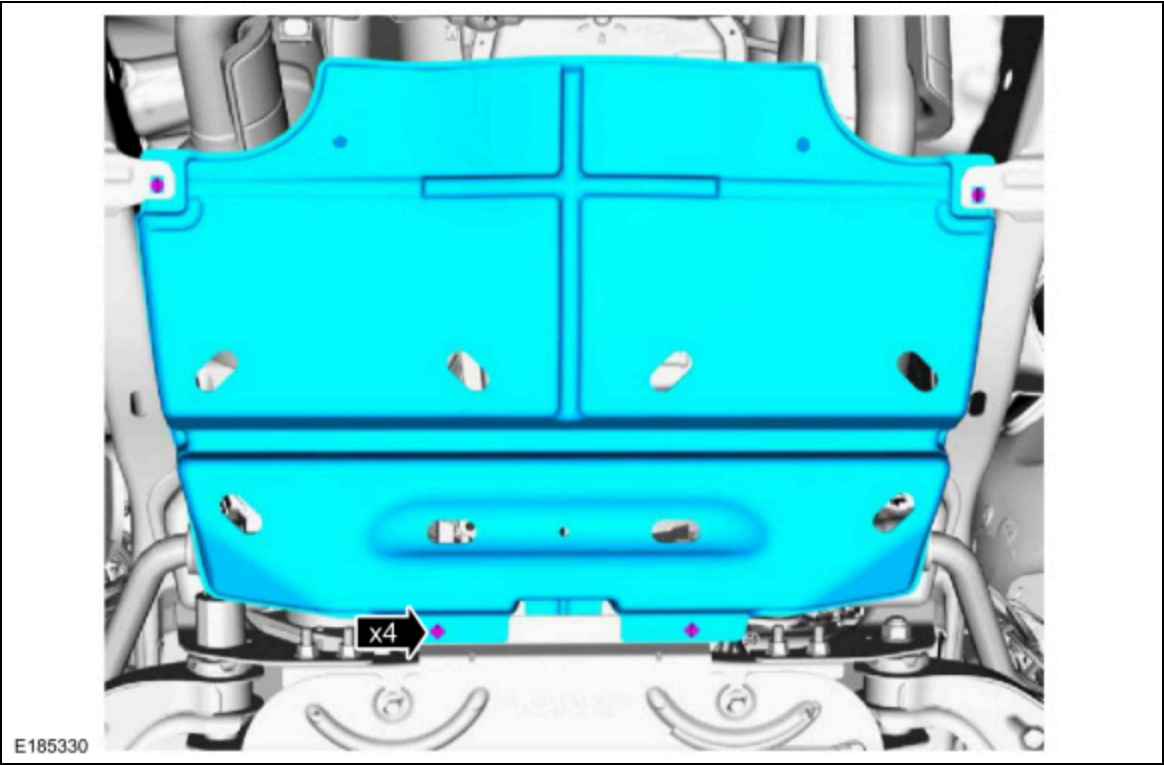
 [Ford Ecat Electronic Parts Catalog](#)

Material	
Item	Specification
Motorcraft® High Temperature Nickel Anti-Seize Lubricant XL-2	—

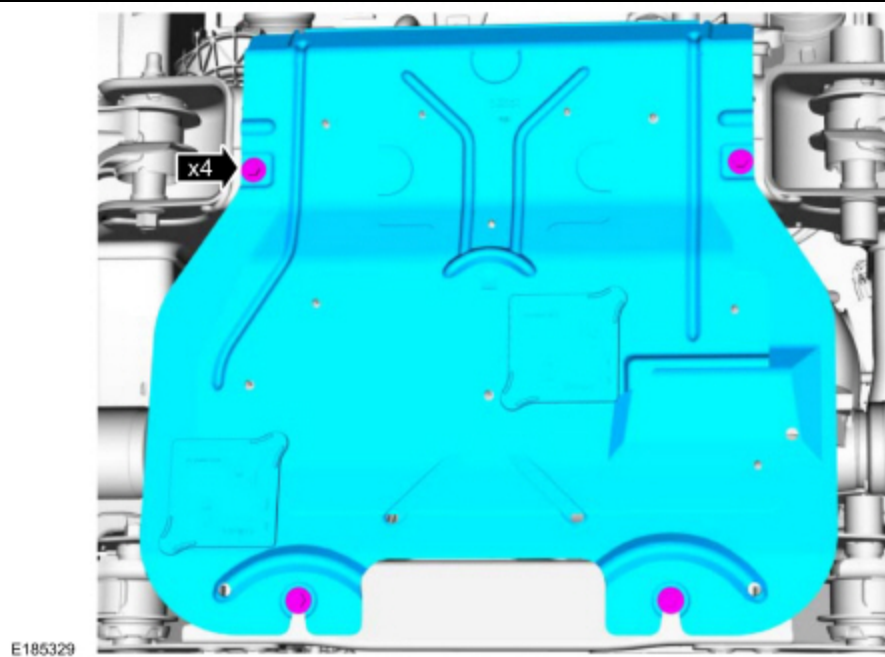
Removal

NOTICE: Whenever turbocharger air intake system components are removed, always cover open ports to protect from debris. It is important that no foreign material enter the system. The turbocharger compressor blades are susceptible to damage from even small particles. All components should be inspected and cleaned, if necessary, prior to installation or reassembly.

1. With the vehicle in NEUTRAL, position it on a hoist, refer to Section 100-02.
2. Drain the cooling system, refer to Section 303-03.
3. Release the underbody splash shield fasteners and remove the underbody splash shield.



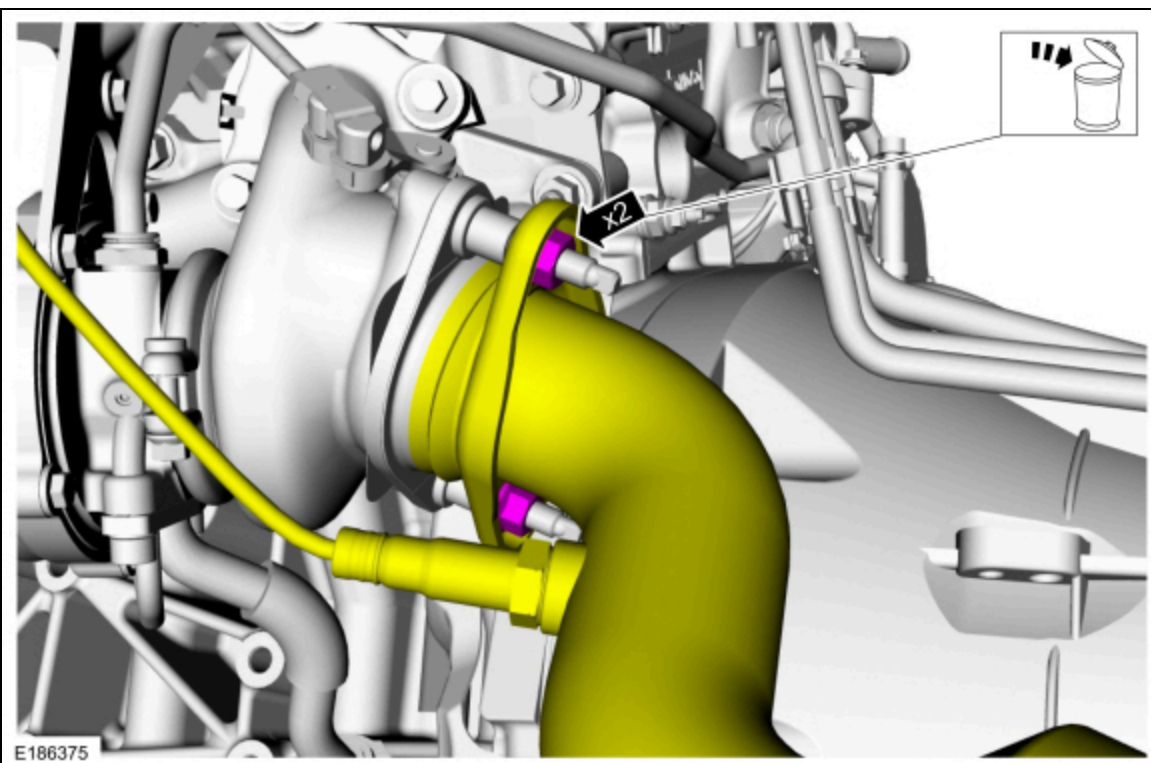
4. If equipped, remove the skid plate bolts and remove the skid plate.



5. Remove the **L.F.** Fender Splash Shield, refer to Section 501-02.

6. Remove the Air Cleaner (ACL) outlet pipe, **L.H.** ACL pipe-to-turbocharger pipe and the **L.H.** turbocharger-to-Charge Air Cooler (CAC) pipe. Refer to Intake Air System Components — Exploded View, 3.5L Gasoline Turbocharged Direct Injection (GTDI) and Air Cleaner Outlet Pipe — Gasoline Turbocharged Direct Injection (GTDI) in Section 303-12.

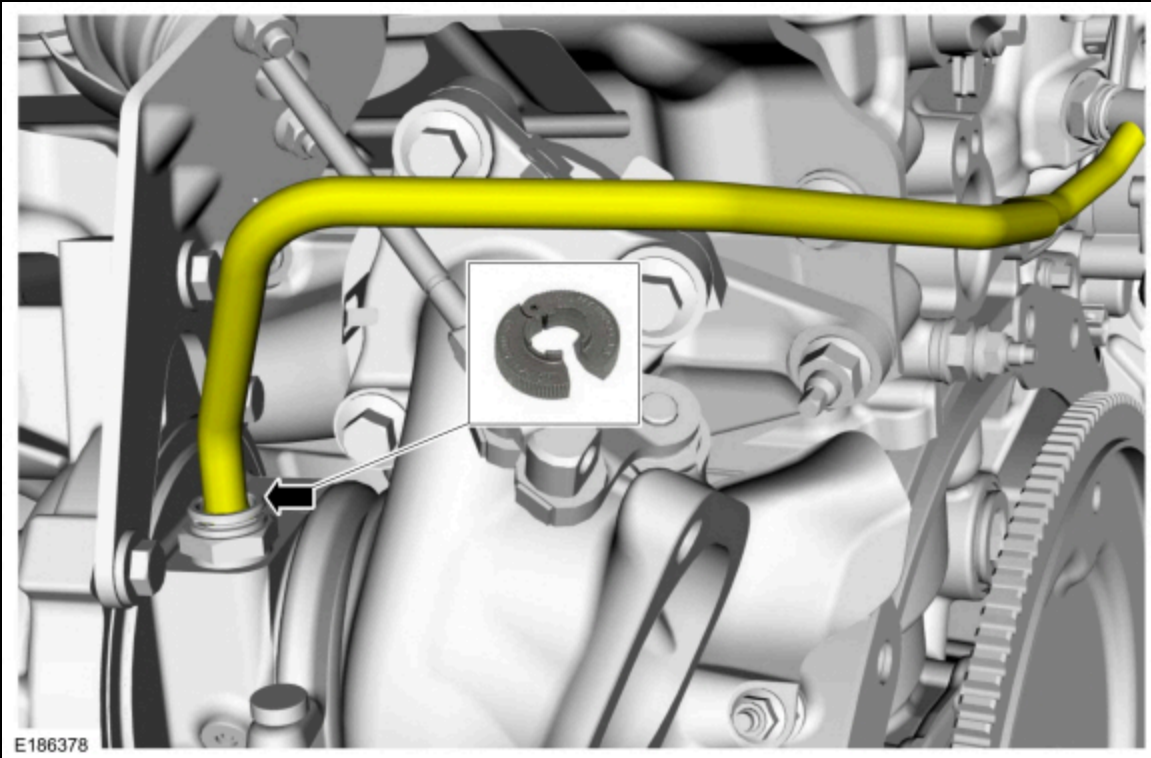
7. Remove and discard the catalytic converter adapter nuts.



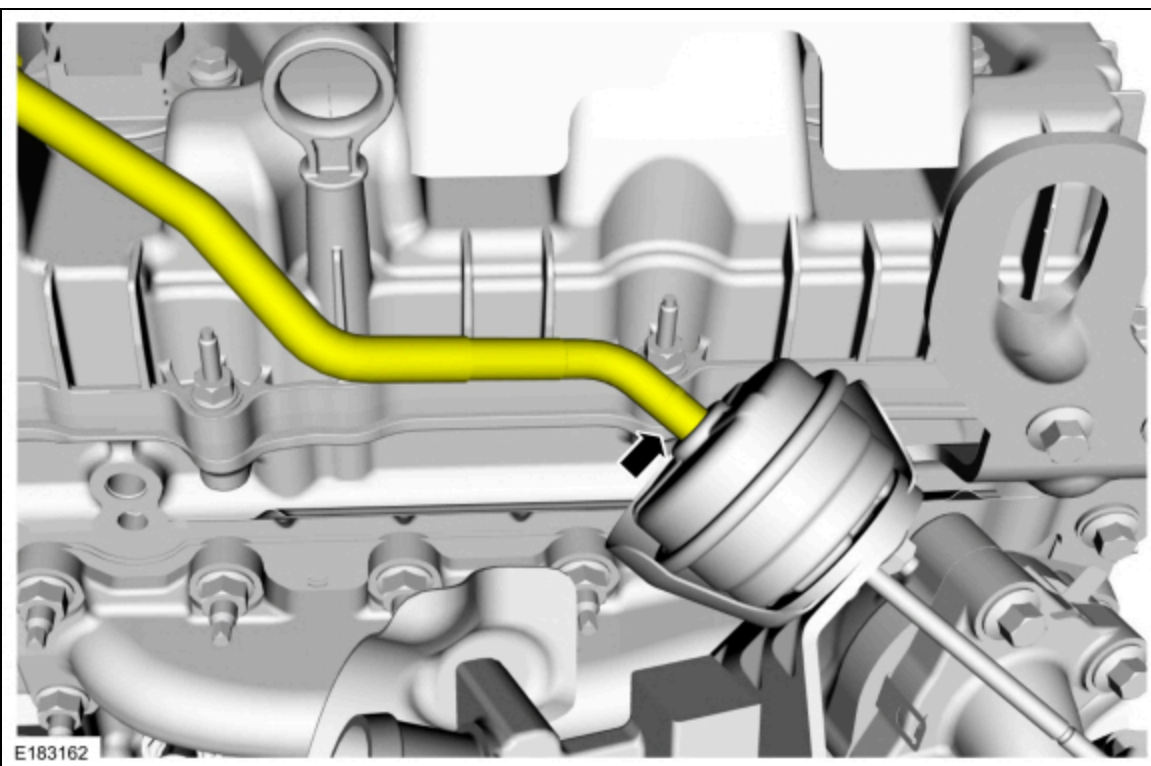
8. **NOTE:** Position a drain pan prior to disconnecting the turbocharger coolant supply tube.

NOTE: Use a 3/8" Jiffy-Tite® quick disconnect tool, close the tool around the tube, press the tool into the fitting and rotate until the tool is fully seated into the fitting.

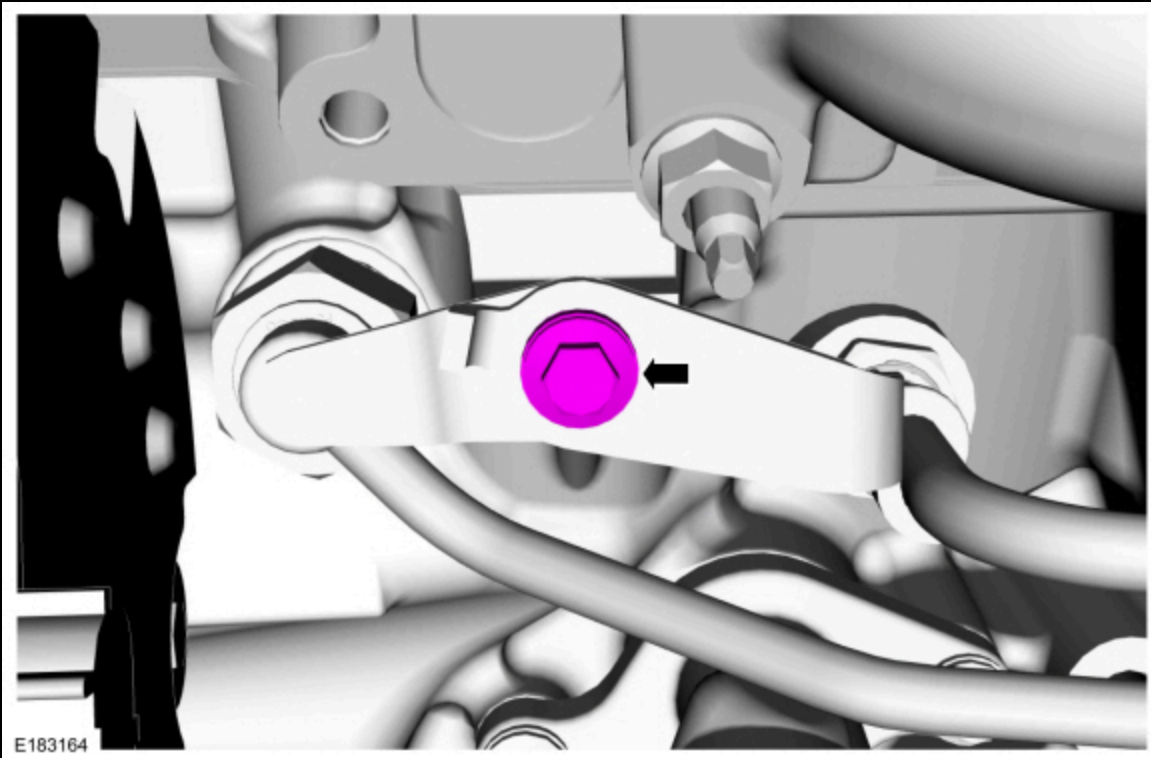
Disconnect the turbocharger coolant return tube at the turbocharger and position the turbocharger coolant return tube up out of the way.



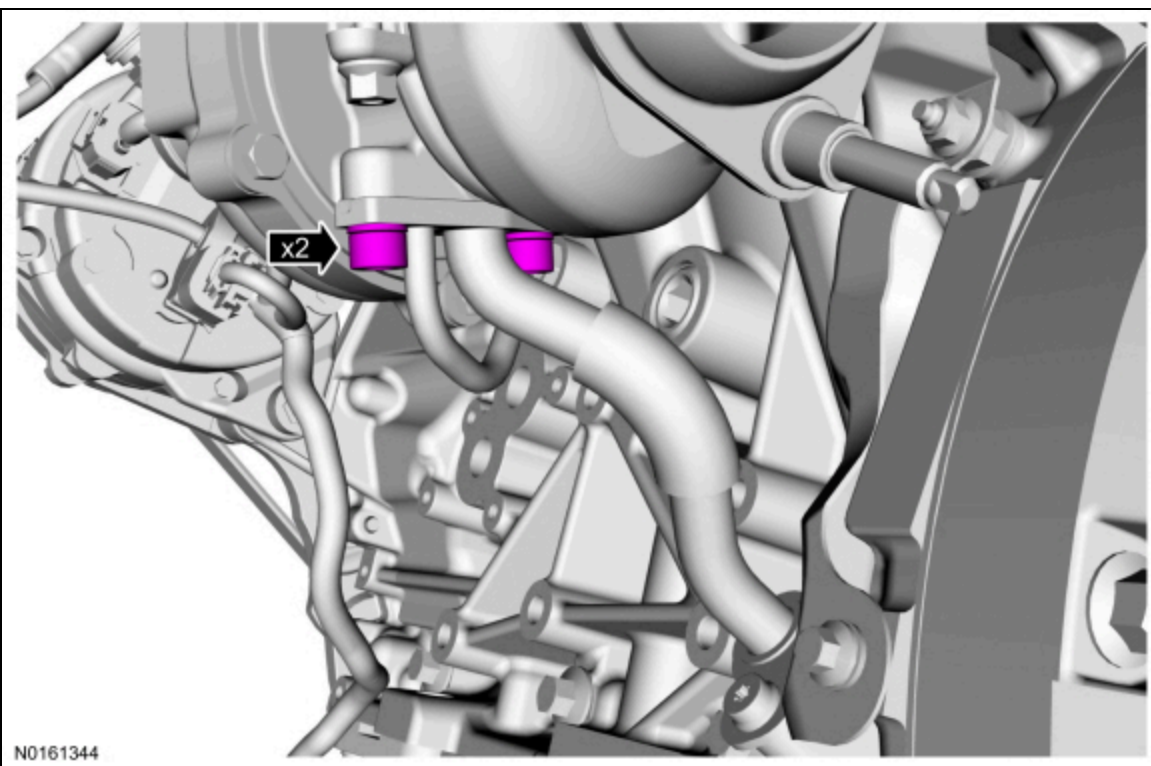
9. Disconnect the turbocharger wastegate activator hose.



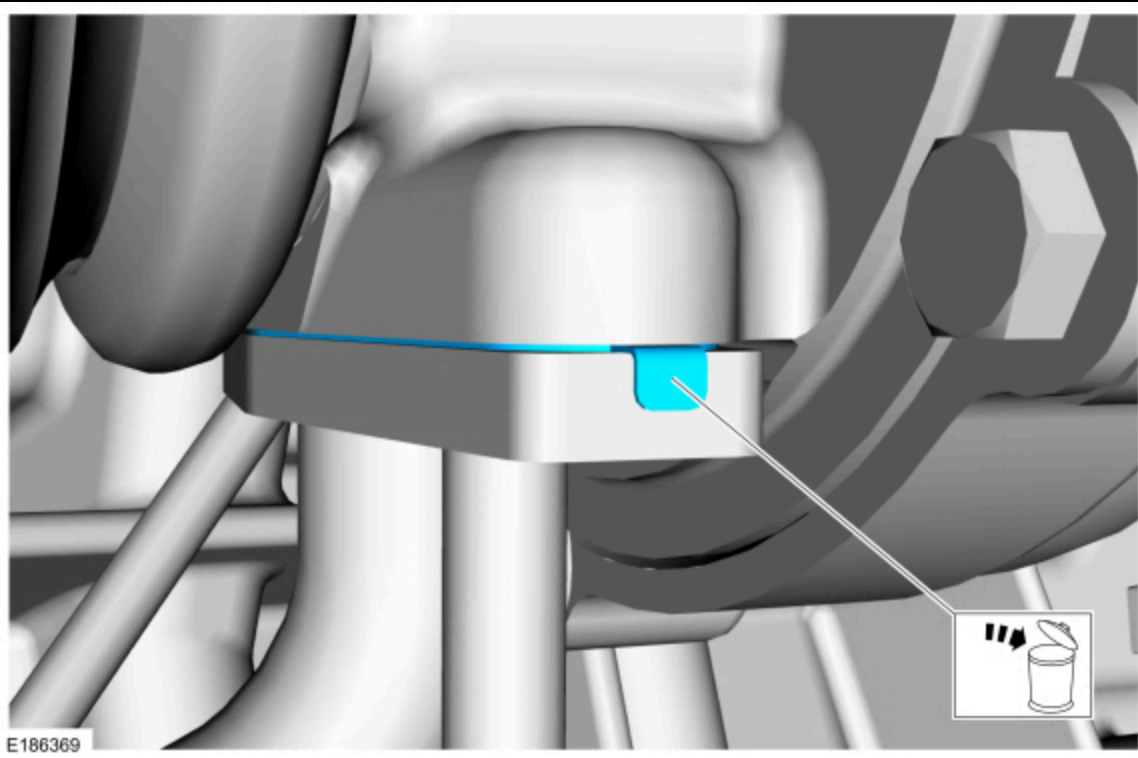
10. Remove the turbocharger coolant supply tube and the turbocharger oil feed tube bolt.



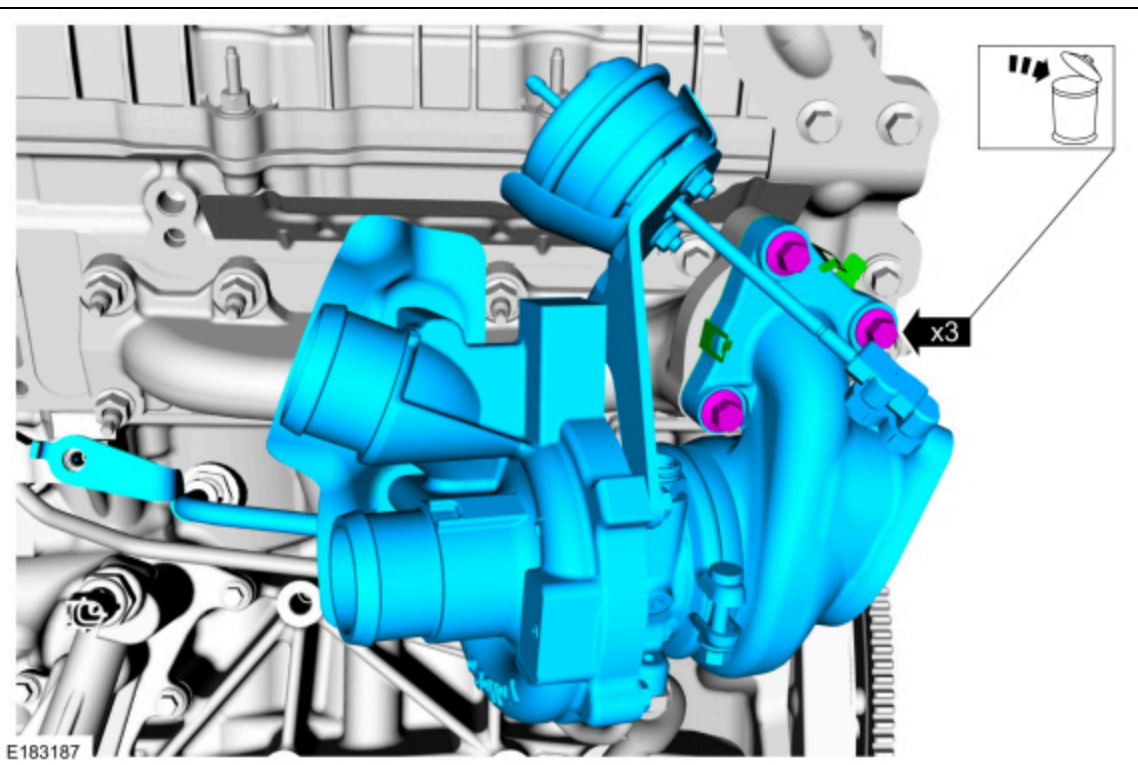
11. Remove the upper turbocharger oil tube assembly bolts at the turbocharger.



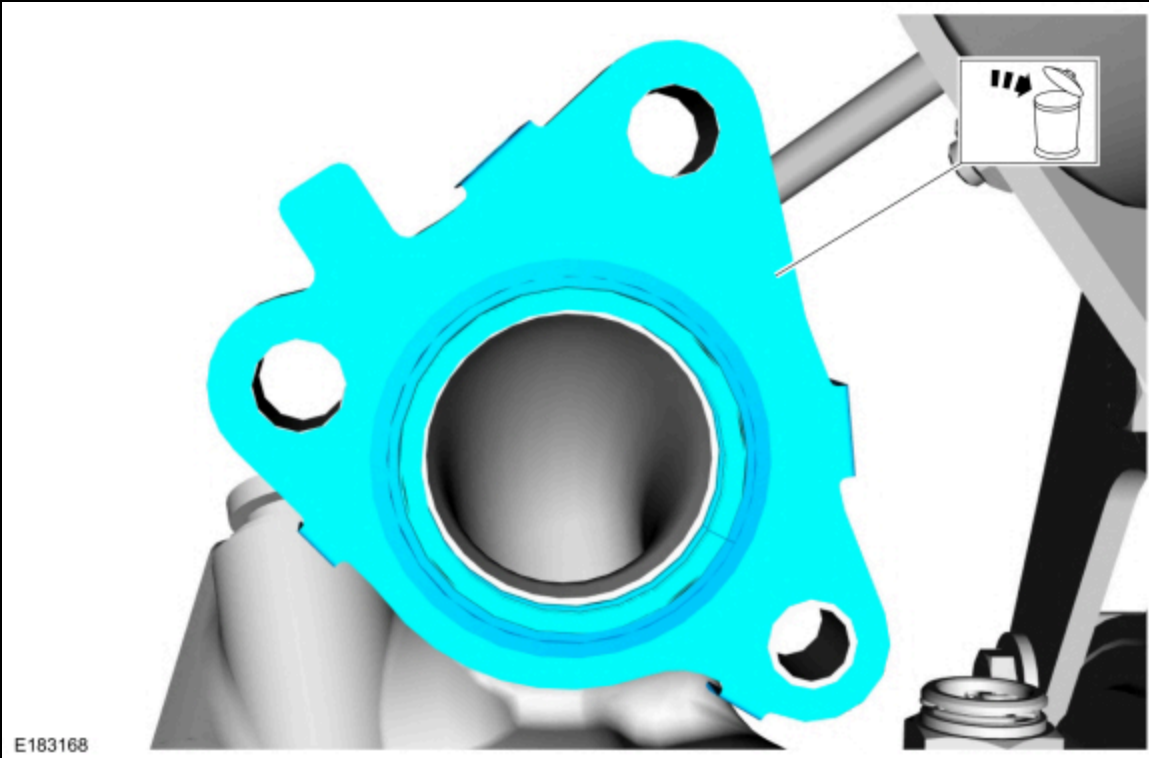
12. Remove and discard the turbocharger oil tube assembly gasket.



13. Remove and discard the turbocharger mounting bolts, remove the turbocharger coolant supply tube from the block fitting while removing the turbocharger.

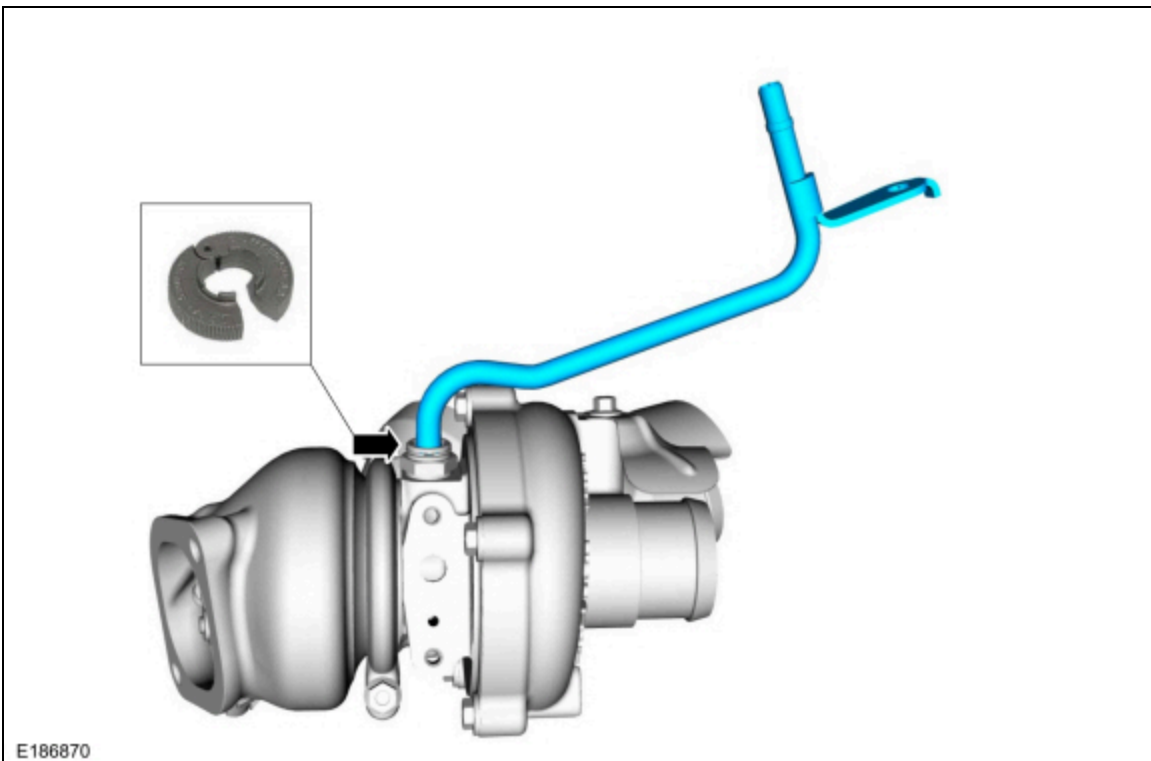


14. Remove and discard the turbocharger gasket.

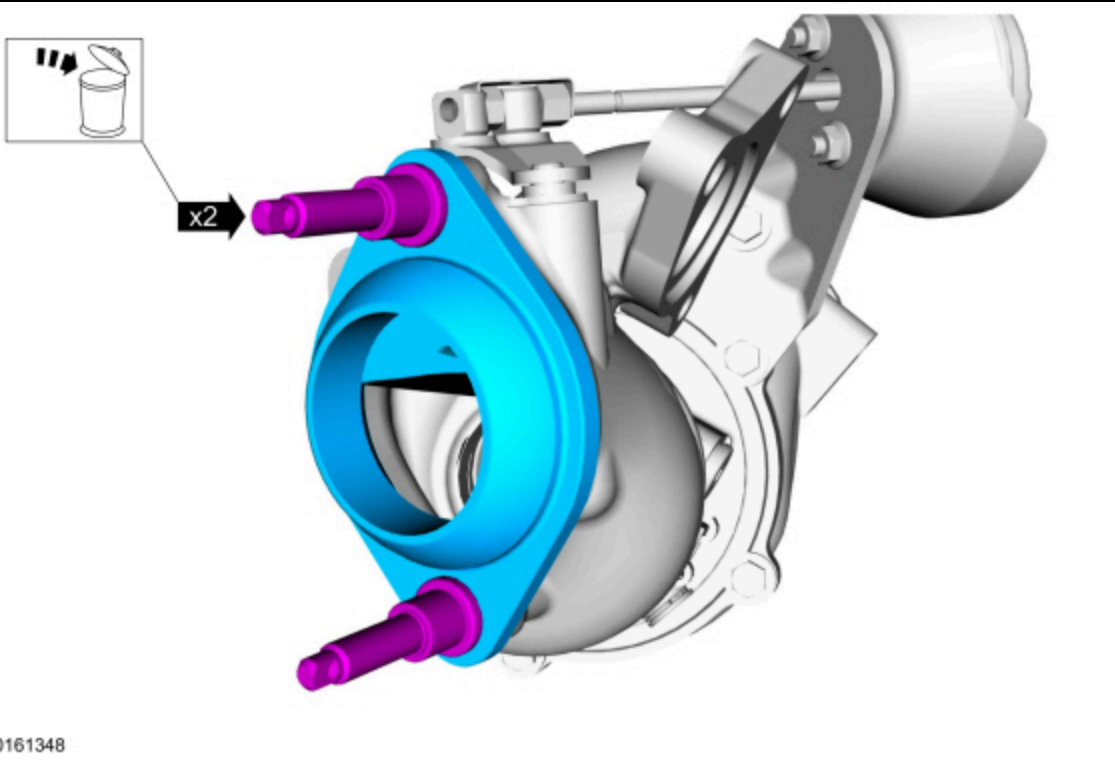


15. **NOTE:** Use a 3/8" Jiffy-Tite® quick disconnect tool, close the tool around the tube, press the tool into the fitting and rotate until the tool is fully seated into the fitting.

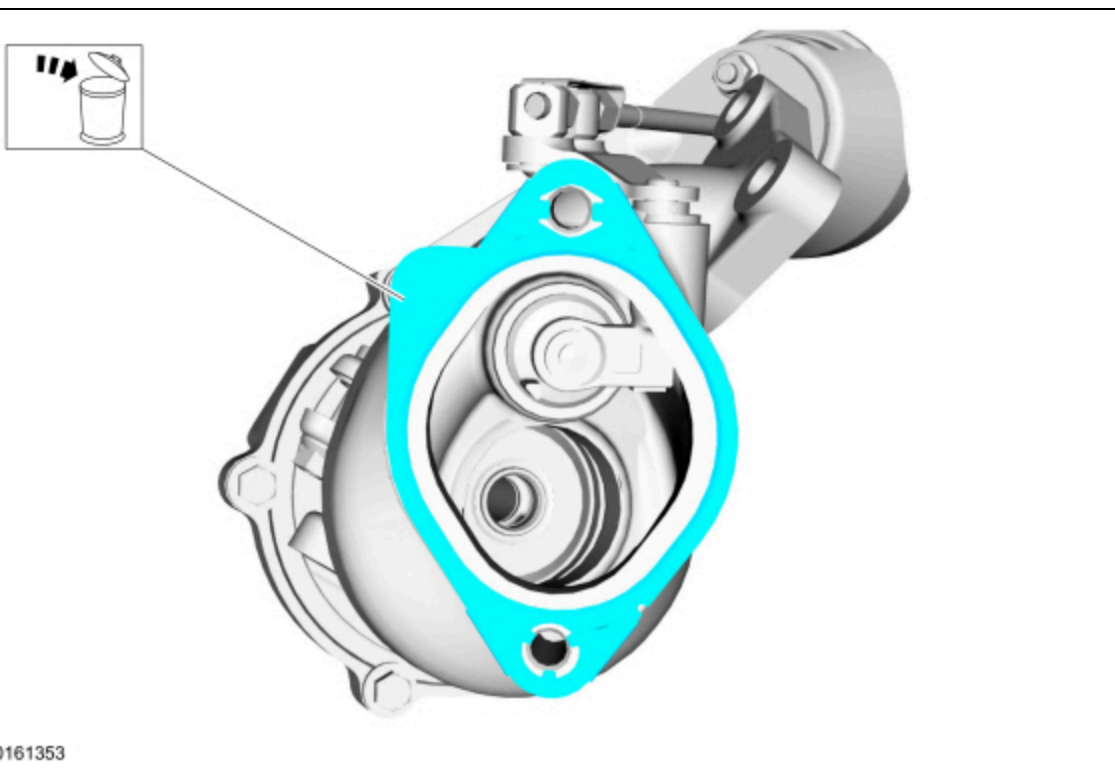
If necessary, disconnect the turbocharger coolant supply tube from the turbocharger.



16. Remove and discard the turbocharger exhaust adapter stud bolts and then remove the turbocharger exhaust adapter.



17. Remove and discard the turbocharger exhaust adapter gasket.



18. **NOTE:** *Whenever the turbocharger is removed or replaced, always install a new turbocharger fitting and oil filter assembly.*

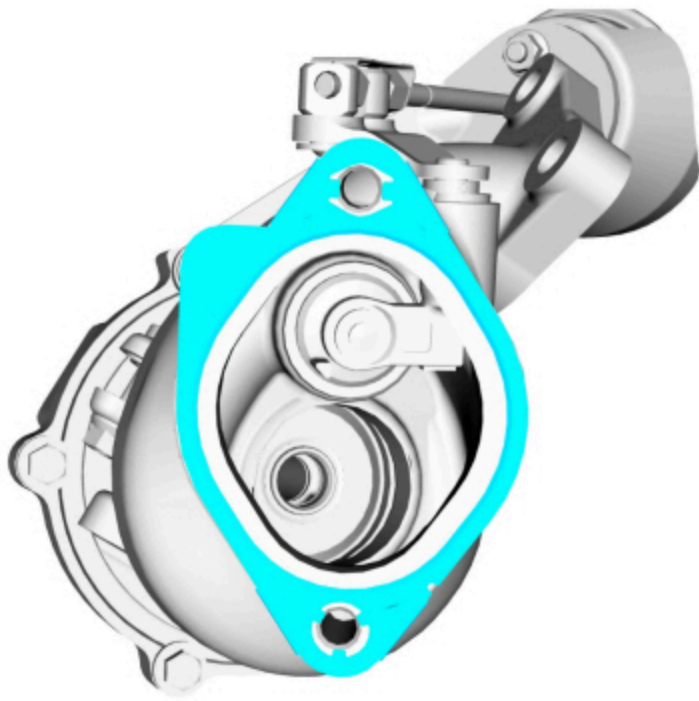
Remove the turbocharger oil supply tube **LH** .
Refer to: Turbocharger Oil Supply Tube — LH.

Installation

1. **NOTE:** *Whenever the turbocharger is removed or replaced, always install a new turbocharger fitting and oil filter assembly.*

Install the turbocharger oil supply tube **LH** .
Refer to: Turbocharger Oil Supply Tube — LH.

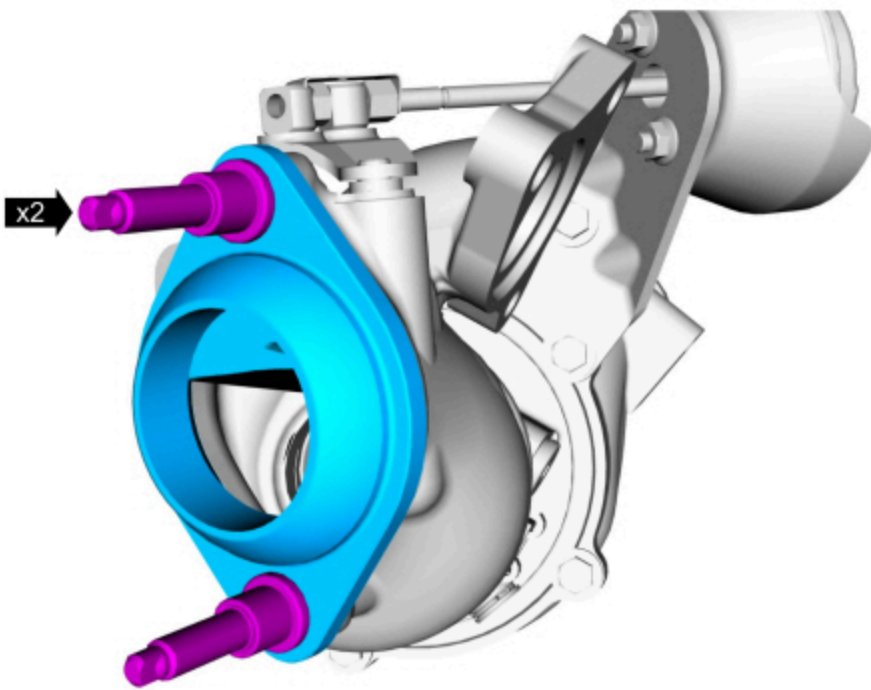
2. Install the new the turbocharger exhaust adapter gasket.



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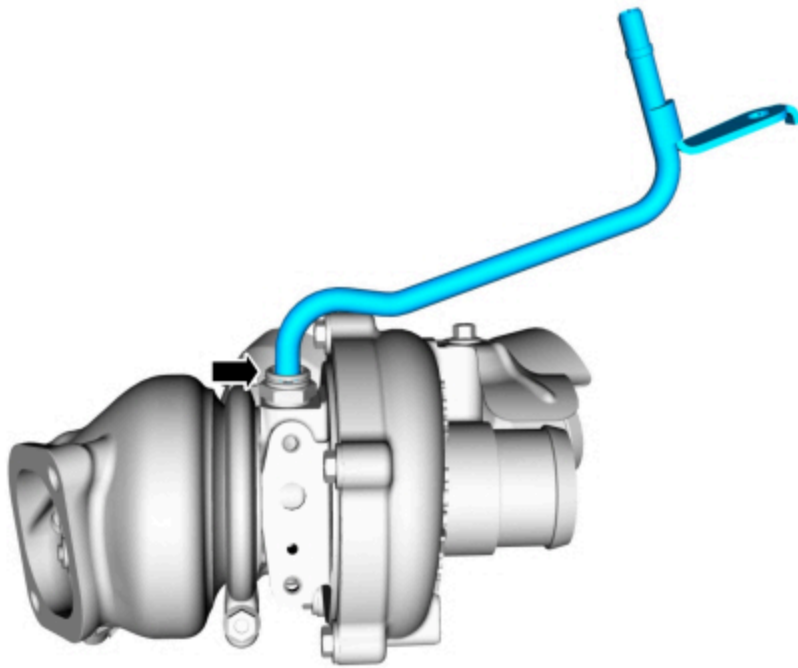
3. **NOTE:** Apply high temperature nickel anti-seize lubricant to the turbocharger exhaust adapter stud bolt threads prior to the installation of the turbocharger exhaust adapter. Material: Motorcraft® High Temperature Nickel Anti-Seize Lubricant XL-2.

Install the turbocharger exhaust adapter and the new turbocharger exhaust adapter stud bolts and tighten to 40 Nm (30 lb-ft).



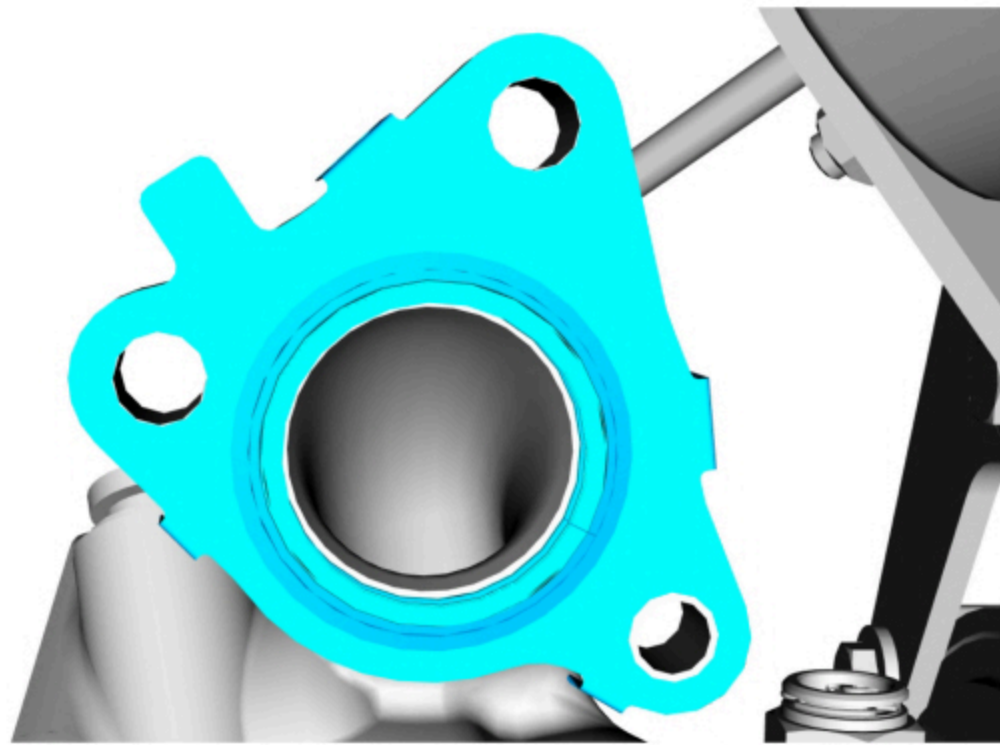
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4. If removed, install the turbocharger coolant supply tube.



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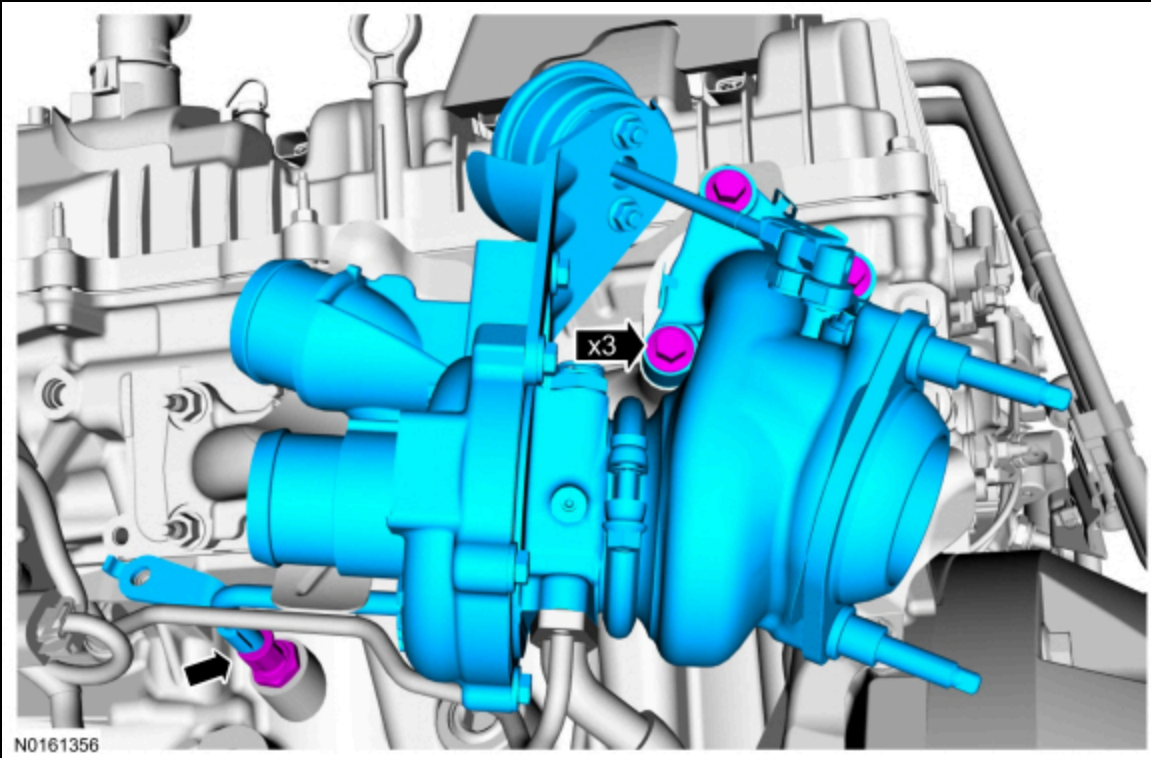
5. Install the new turbocharger gasket.



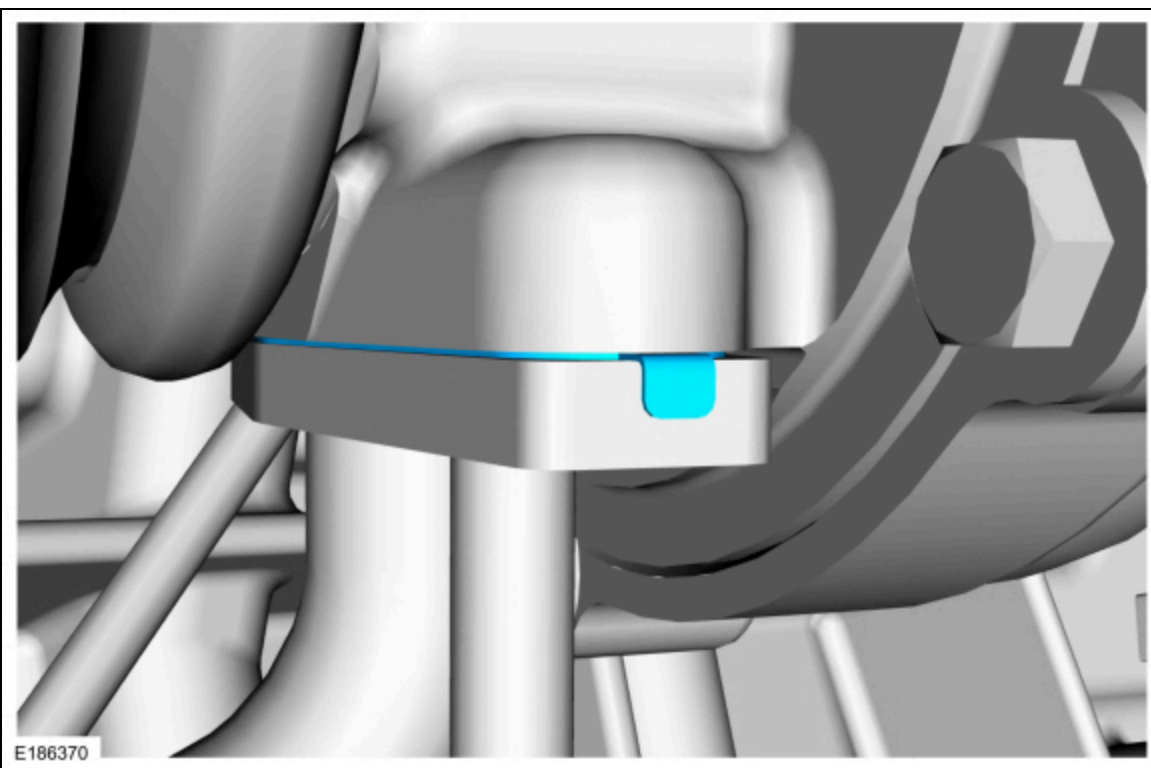
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6. **NOTE:** *Align catalytic converter pipe flange to the turbocharger adapter while installing the turbocharger.*

Install the new turbocharger mounting bolts, install the turbocharger coolant supply tube to the block fitting while installing the turbocharger. Tighten the new turbocharger mounting bolts to 32 Nm (24 lb-ft).

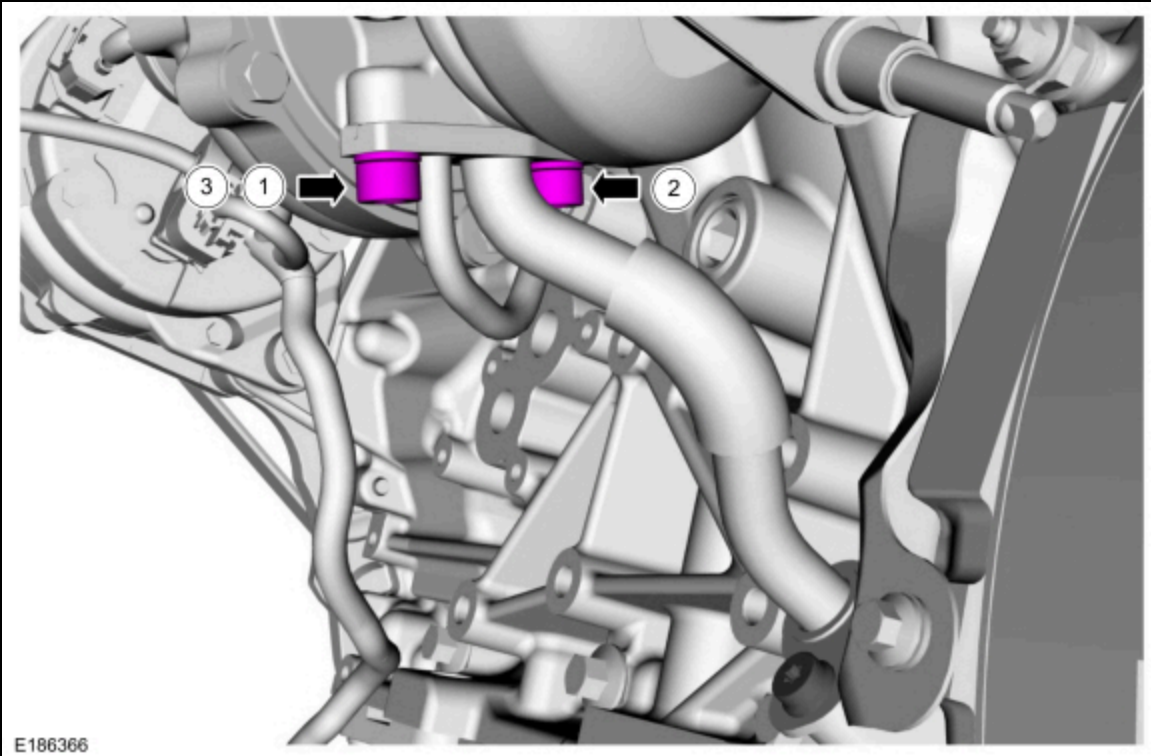


7. Install the new turbocharger oil tube assembly gasket.



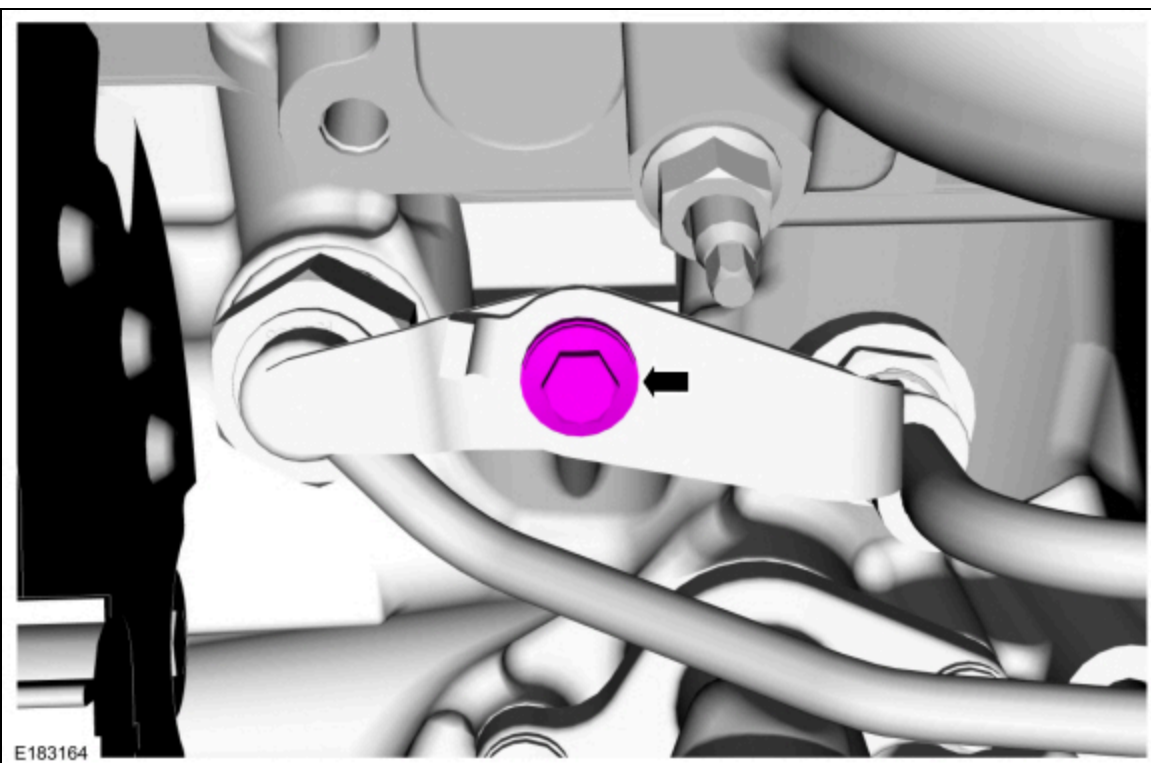
8. Install the turbocharger oil tube assembly upper bolts finger tight at the turbocharger.

1. Tighten the outer turbocharger oil tube bolt to 5 Nm (44 lb-in).
2. Tighten the inner turbocharger oil tube bolt in the following 2 stages.
 - Stage 1: Tighten the inner turbocharger oil tube bolt to 10 Nm (89 lb-in).
 - Stage 2: Tighten the inner turbocharger oil tube bolt an additional 30 degrees.
3. Tighten the outer turbocharger oil tube bolt in the following 2 stages.
 - Stage 1: Tighten the outer turbocharger oil tube bolt to 10 Nm (89 lb-in).
 - Stage 2: Tighten the outer turbocharger oil tube bolt an additional 30 degrees.

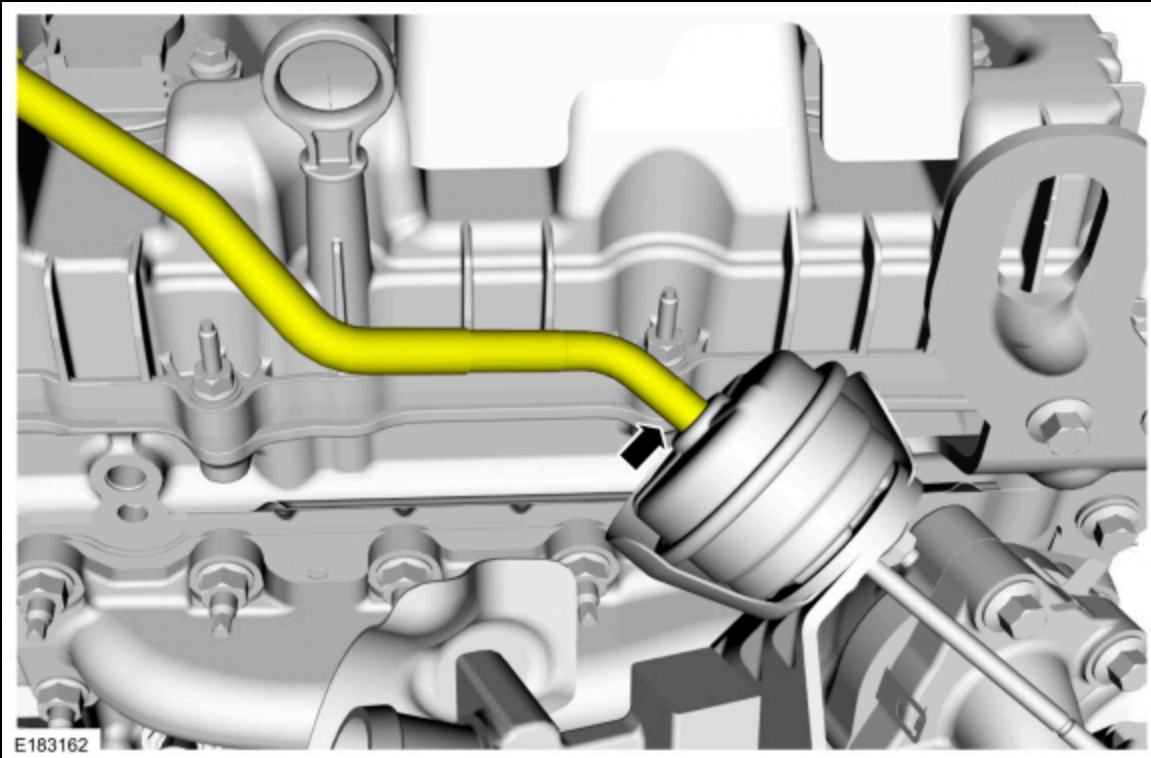


9. Install the turbocharger coolant supply tube and the turbocharger oil feed tube bolt. Tighten the turbocharger coolant supply tube and the turbocharger oil feed tube bolt in the following 2 stages.

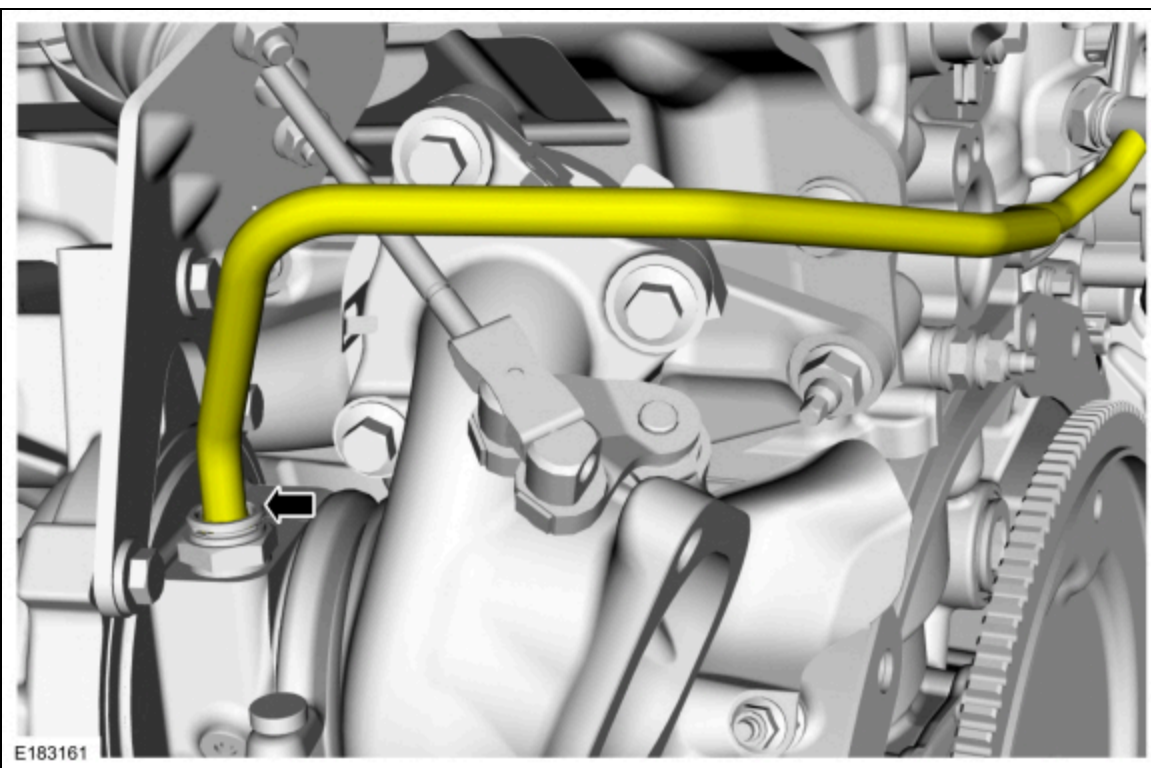
- Stage 1: Tighten to 8 Nm (71 lb-in).
- Stage 2: Tighten an additional 30 degrees.



10. Connect the turbocharger wastegate activator hose.

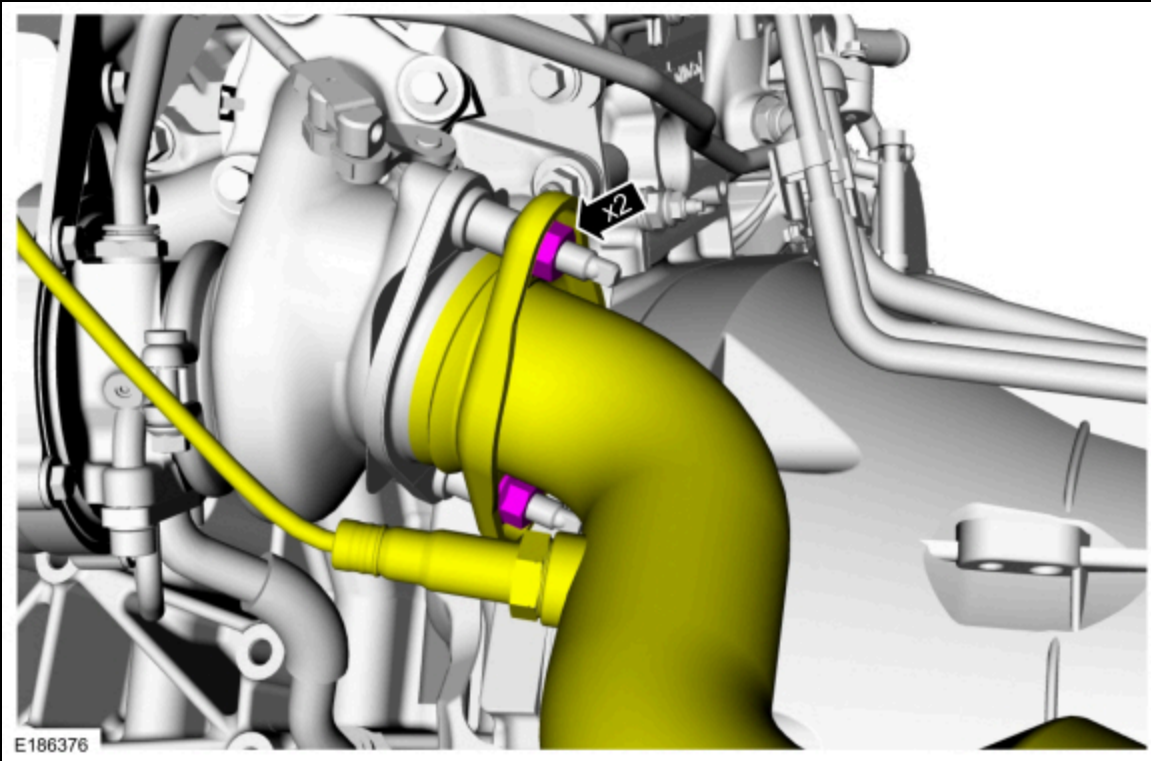


11. Connect the turbocharger coolant return tube to the turbocharger.

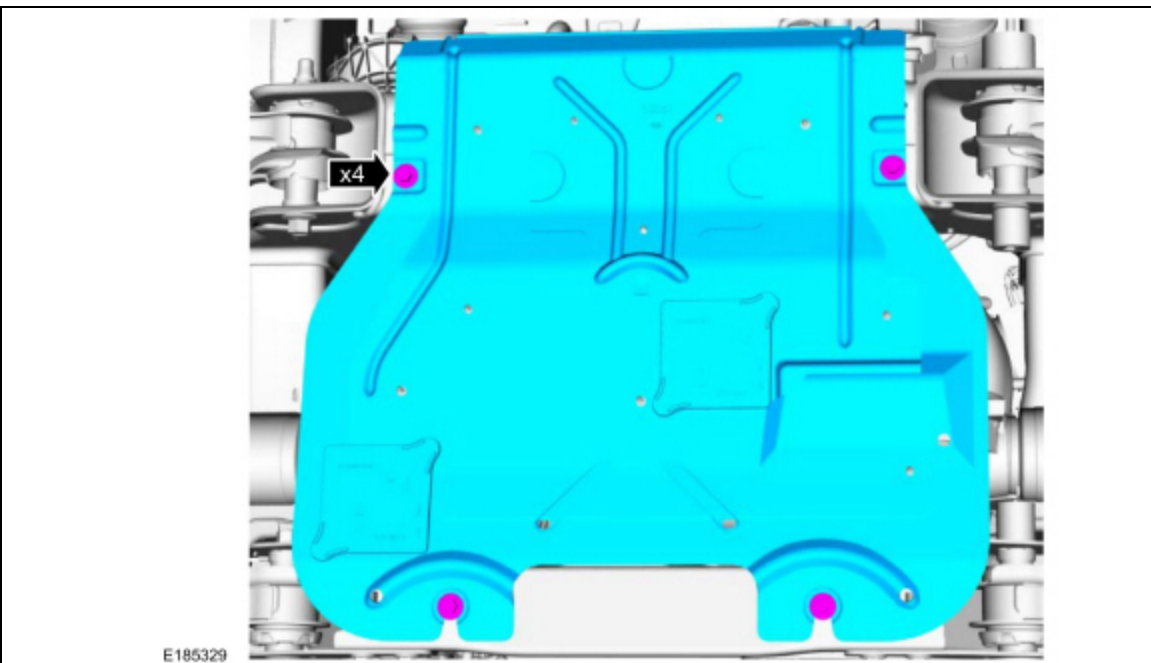


12. **NOTE:** *Install the new catalytic converter adapter nuts finger tight.*

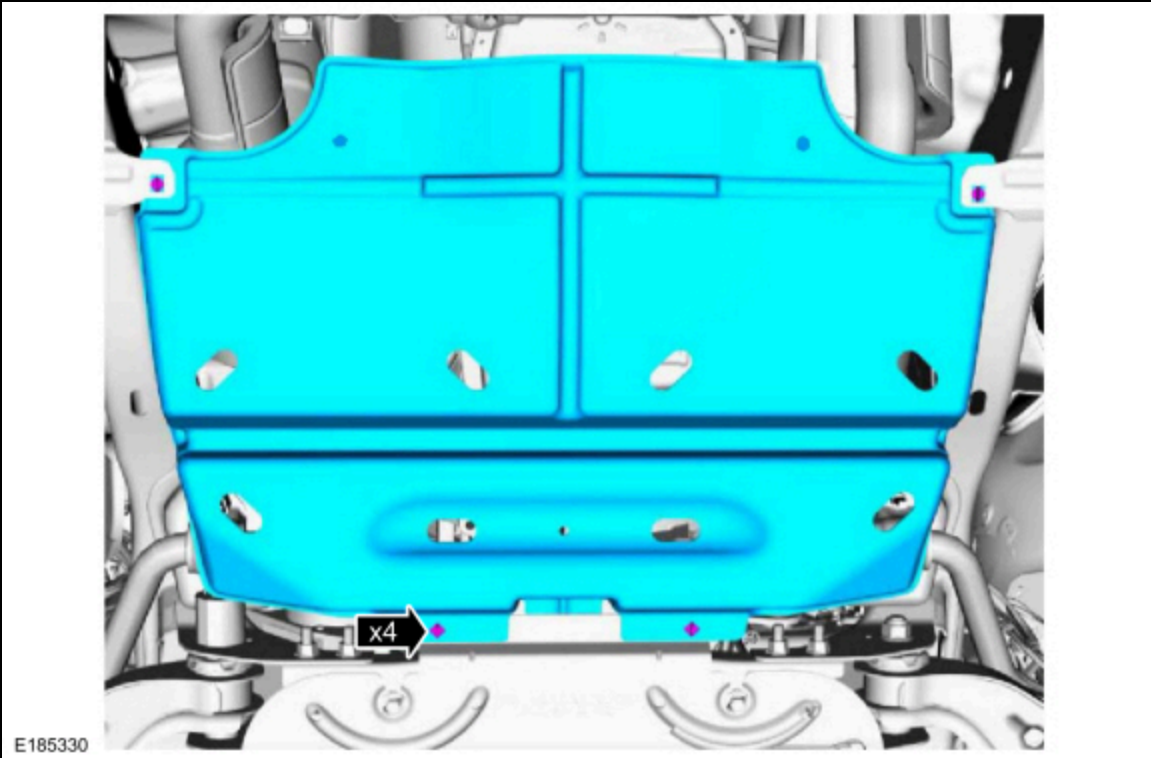
Evenly tighten the new catalytic converter adapter nuts to 40 Nm (30 lb-ft).



13. Install the ACL outlet pipe, LH ACL pipe to-turbocharger pipe and the LH turbocharger-to-CAC pipe. Refer to Intake Air System Components — Exploded View, 3.5L GTDI and Air Cleaner Outlet Pipe — Gasoline Turbocharged Direct Injection (GTDI) in Section 303-12.
14. Install the LF Fender Splash Shield, refer to Section 501-02.
15. If equipped, install the skid plate and the skid plate bolts and tighten to 40 Nm (30 lb-ft).



16. Install the underbody splash shield and the underbody splash shield retainers.



17. Fill and bleed the cooling system, refer to Section 303-03.